

Equivariance in Category Theory

Zih-Yu Hsieh

August 4, 2026

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1 Motivation

To study some “things”, what’s a good way of studying it? One thing to do, is studying some other things that’re interconnected to it.

For instance, in topological K-theory, people study vector bundles over a space X , and turn them into sequence of spaces called “spectrum” associated to X . This sequence of spaces and their topologies gives some interesting invariants of the space itself.

Similarly, in algebraic K-theory, people study “projective modules” over a ring R , we can also construct spectrum over this ring, and the topologies also give some interesting invariants of the ring.

However, for both of these cases (vector bundles and projective modules), they’re both closed under direct sums and tensor products (whatever that is), so we actually starts with a “category” with some nice product structure. These are called symmetric monoidal categories. In general, a K-theory can be done, by turning some symmetric monoidal category into a spectrum. So, classifying them turns out to be extremely important.

Now, analogous constructions can be done if the “things” endow some kind of group actions, which is the main study of equivariant K-theory. For this, an important question must be asked: What the heck is a “Symmetric Monoidal G -Category”? Today, we try to get to this, by first introducing equivariance in categories.

2 G -Categories and 2-Category $\mathbf{Cat}_G$

Fix a group G (say finite).

Definition 2.1. A small category $\mathcal{A}$ is a (strict) G -category, if there exists a map $H : G \rightarrow \mathbf{Cat}(\mathcal{A}, \mathcal{A})$, such that $H(e) = \mathrm{id}_{\mathcal{A}}$, and $H(gh) = H(g) \circ H(h)$.

So, each group element is a functor, and an automorphism on $\mathcal{A}$, where composition functors are compatible with group structures. Just like how it's defined for G -sets.

Definition 2.2. The 2-category $\mathbf{Cat}_G$ consists of all small G -categories, the 1-morphisms between G -categories are functors (not necessarily equivariant), and the 2-morphisms are natural transformations between them.

An interesting thing to observe: each G -category $\mathcal{A}$ realizes $g \in G$ as an automorphism on $\mathcal{A}$. So, if $F : \mathcal{A} \rightarrow \mathcal{B}$ is a functor, so is $g \circ F \circ g^{-1}$! This realizes the “functor category” $\mathbf{Cat}_G(\mathcal{A}, \mathcal{B})$ as a small G -category!

If consider the subcategory fixed by all functors of G , $\mathbf{Cat}_G(\mathcal{A}, \mathcal{B})^G$ gives all the G -equivariant functors from $\mathcal{A}$ to $\mathcal{B}$.

3 Operads for Symmetric Monoidal (G) Categories

We're gonna blackbox whatever an operad is. Yes. (Also, there are many texts by Peter describing this). Anyway, a “slogan” for operad, is it encodes how to define n -fold products on some “things”, and how to combine these products.

In $\mathbf{Cat}_G$, there is a “product operation” of G -categories $\mathcal{A} \times \mathcal{B}$, and G acts diagonally on it, similar to G -sets. There are two operads in $\mathbf{Cat}_G$, called $\mathcal{P}$ and $\mathcal{P}_G$, both encodes some methods of describing n -fold products $\mathcal{A}^n \rightarrow \mathcal{A}$ on some G -category $\mathcal{A}$.